A Secure Approach to Mitigate Double Voting in Electronic Voting Systems using Blockchain Technology

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***Abstract*—The Voting DApp is a decentralized application created to offer a safe, open, and impenetrable platform for blockchain-based election administration. A smart contract running on the Ethereum blockchain manages the system's basic voting logic, guaranteeing integrity through secure voter registration, role-based access control, and the avoidance of duplicate voting. Through a React-based user interface that is integrated with Web3.js, registered voters may cast their ballots, and administrators can add candidates and manage voter registrations. The Voting DApp is a strong solution for contemporary digital voting systems since it makes use of blockchain's immutability and transparency to guarantee electoral process trust and offer real-time result visibility.**

Keywords*—Voter Registration, Role based Access, MetaMask, Blockchain, Decentralized Application, Candidate Management.

I. Introduction

The need for safe, open, and effective voting procedures has grown in importance in the current digital era. Traditional voting methods are frequently beset by issues of lack of transparency, fraud, and meddling. In order to overcome these obstacles, the Voting DApp uses block chain technology to build a decentralized platform that guarantees electoral security, integrity, and confidence. This method offers a transparent and impenetrable solution for contemporary elections by removing the hazards connected with centralized voting systems through the employment of smart contracts and the Ethereum block chain.

A smart contract and an intuitive React-based front end work together to power the Voting DApp. The core of the application is the smart contract, which enforces regulations including candidate management, voter registration, and duplicate voting avoidance. Role-based access is one of the main features; only the administrator can add candidates and register voters, and voters can safely cast their ballots following verification. Voter confidence is increased by the system's decentralized structure, which guarantees that all voting records are public, auditable, and unchangeable. The Voting DApp allows users and the Ethereum network to communicate seamlessly by integrating Web3.js and blockchain technologies. To take part in elections, voters can link their MetaMask wallets, which guarantees safe authentication. The smart contract's events give the front-end interface real-time changes, enabling users to watch election results in full transparency. In addition to addressing the shortcomings of conventional voting methods, this initiative opens the door for easily accessible and expandable digital voting solutions in a number of industries.

II. Review of Existing Work

The establishment of trust and transparency in decentralized systems has been completely transformed by block chain technology, especially in domains where security and integrity are crucial, like voting. By introducing block chain technology with Bitcoin, Authors [1] laid the foundation for smart contracts and decentralized apps (DApps). Building upon this framework, Ethereum became a prominent DApp platform with strong programmable smart contract support [2]. Long-standing issues with traditional voting methods, like fraud, tampering, and lack of transparency, have been addressed by these advancements, which have made it easier to create safe and impenetrable systems.

The employment of block chain technology in voting systems has been the subject of numerous research. Authors [3] emphasized how blockchain may be used to guarantee security and transparency in Internet of Things (IoT) applications, which is in line with what contemporary voting systems need. In the study [4] of Ethereum smart contract vulnerabilities, emphasized the significance of secure coding techniques in thwarting assaults. Furthermore, authors [5] investigated formal verification of smart contracts, showing how these methods can improve voting systems by guaranteeing the accuracy of crucial features including results calculation, voter registration, and the avoidance of duplicate

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voting. The difficulties of user interaction with blockchain systems are also being addressed by recent research [6-8]. The integration of Web3.js and React for smooth front-end interaction with block chain backend was covered by authors [9-11]. Non-technical individuals can safely interact with decentralized systems thanks to these technologies. Additionally, research on block chain-based identity verification, such that done by authors [12], offers guidance on how to put strong voter authentication into practice. As the Voting DApp project demonstrates, these efforts collectively provide the framework for the creation of safe [13-16], open, and intuitive blockchain-based voting systems.

The literature review [17-18] emphasizes how block chain technology has the ability to revolutionize decentralized systems like voting by improving security, transparency, and trust. Authors [19] laid the groundwork for block chain and Ethereum as the main components for developing decentralized, tamper-proof apps. The application of smart contracts in voting has been studied, with a focus on how their secure coding and formal verification might guard against fraud and guarantee integrity. Research also tackles [20] user interface issues, highlighting front-end integration solutions such as Web3.js and React. Additionally, voter authentication is strengthened by developments in identity verification, which makes block chain-based voting systems safe, transparent, and usable for contemporary electoral requirements.

III. Research Gap and Objectives

Lack of transparency, fraud susceptibility, ineffective vote counting, and difficulties protecting voter security and privacy are only a few of the major issues facing traditional voting methods. The integrity of democratic processes and public trust are threatened by these problems. Even if they address certain issues, current electronic voting methods frequently fail to provide elections that are verifiable and impenetrable. A decentralized, safe, and transparent voting process is made possible by the development of block chain technology, which presents a promising answer to these issues. In order to provide a safe and effective platform for holding elections, this project intends to create a block chain based online voting system that addresses issues of scalability and usability while guaranteeing integrity, anonymity, and transparency.

A. Objectives

- Increase the Transparency of the Voting Process

The objective of the work is to inculcate a transparent voting system using block chain technology. Every vote is securely recorded, guarding against any illegal changes, thanks to block chain's decentralized and immutable nature. This transparency offers a contemporary response to long-standing worries about election credibility and accountability by working to dispel suspicions about electoral integrity and promote confidence in the process among voters and authorities.

- Protect Voter Privacy and Anonymity

One of the project's main goals is to protect voter anonymity. Voter identities will be protected by sophisticated cryptographic mechanisms, guaranteeing that every vote is cast in confidence and cannot be linked to any specific person. This strategy encourages free and fair participation in the election process without fear or bias by shielding voters from any coercion or reprisal.

- Boost Security and Stop Fraud

A key component of this block chain-based voting system is security. The project intends to remove vulnerabilities such as vote tampering, double voting, or illegal access by incorporating strong cryptographic methods. By guaranteeing that every vote is verifiable and impervious to manipulation, block chain's immutable ledger greatly lowers the possibility of electoral fraud and strengthens the legitimacy of the voting procedure.

- Usability and Accessibility

By creating an easy-to-use interface that enables voters to safely cast their ballots from any place, the initiative aims to make voting inclusive. This feature makes it possible for people with physical limitations or those living in remote places to join without facing logistical obstacles. The system guarantees that all qualified voters can easily exercise their democratic rights by fostering accessibility.

- Increase Election Process Trust

Restoring confidence in election systems is one of the main goals. Accurate and equitable results are ensured by the system's decentralized, impenetrable platform. Block chain's transparency and auditability may be relied upon by voters, candidates, and election officials, removing uncertainty over election results and bolstering the democratic framework for all parties involved.

B. Data Acquisition

In order to guarantee safe and effective operation, data acquisition for the block chain-based online voting system entails obtaining and organizing necessary information. In order to prevent fraud, key data includes voter registration details, such as unique IDs and demographic data, which are obtained from government electoral databases. The system is configured using election-specific data, such as constituency boundaries and candidate details. In order to maximize scalability, performance metrics are gathered during testing, such as transaction speeds and block processing times. The confidentiality and integrity of votes are guaranteed by cryptographic parameters that are taken from secure protocols. Pilot test feedback helps identify possible problems and offers insights on usability. The system is built to provide a safe, open, and expandable voting solution by utilizing trustworthy data sources and thorough validation.

- Smart Contracts Design

At the heart of the suggested block chain-based voting system are smart contracts, which are automated, impenetrable algorithms that manage crucial election functions including vote casting, storage, and counting. These contracts guarantee that a voter's vote is safely stored on the block chain and cannot be changed once it is cast. Before enabling a voter to cast a ballot, the smart contract verifies their eligibility using their unique ID or digital signature, among other predetermined requirements. Furthermore, it removes the need for centralized middlemen by enabling real-time vote tallying while protecting voter confidentiality. By eliminating the possibility of human error or tampering, this automation improves the voting process's efficiency, accuracy, and transparency. Strict testing along with auditing protocols are used to assure the security and dependability of smart contracts. Formal verification is performed on the

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contract logic to look for flaws such as denial of service problems, overflow mistakes, and reentrancy assaults. To verify that the system works in a variety of situations, simulated elections are held in controlled settings. External audits conducted by block chain security companies supplement these procedures by attesting to the robustness and integrity of the contract code. The technology ensures voter trust and protects the democratic process from possible attacks by following best practices in secure smart contract development.

- Consensus Mechanism

A strong consensus algorithm is used by the block chain based voting system to verify transactions and guarantee the integrity of the voting procedure. How network users agree to add a new vote to the block chain depends on the consensus mechanism they have chosen, such as Proof of Work or Proof of Stake. PoW is extremely safe against manipulation but resource-intensive because it depends on computational effort to solve intricate mathematical puzzles. As an alternative, PoS offers a more scalable and energy-efficient option by choosing validators according to their stake in the network. The selected technique in this system guarantees that votes are precisely recorded and unchangeable, promoting openness and confidence among voters. Because it takes a lot of computing power or resources to launch an attack, it also stops bad actors from tampering with results.

The chosen method is justified by comparison with alternative consensus techniques. For instance, by depending on a small number of elected validators, Delegated Proof of Stake (DPoS) may increase the risk of centralization even while it speeds up transactions. Similar to this, decentralization may be jeopardized by systems like Proof of Authority (PoA), which provide high efficiency but rely on reliable authorities. The selected approach is perfect for electronic voting because it strikes a balance between security, decentralization, and scalability. This guarantees that the voting procedure is both safe and effective, able to manage large transaction volumes without compromising the accuracy of the outcomes. A thorough assessment of the system's needs for scalability, dependability, and transparency led to the choice of the particular consensus process.

IV. Proposed Methodology

Voter registration, candidate addition, and vote casting are just a few of the crucial voting system operations that smart contracts automate. Developed using Solidity, an Ethereum smart contract programming language, they enable the system to function independently after it is put into use. In order to maintain fairness and eliminate fraud, smart contracts manage tasks like vote validation and vote storage. They eliminate the need for physical intervention by enforcing predetermined conditions and running automatically when the requirements are satisfied. This project uses MetaMask, a bitcoin wallet and gateway to block chain apps, to verify users. In order to provide safe and smooth interactions between the DApp and the block chain, MetaMask offers the interface for connecting the user's wallet to the block chain. Voters can safely register on the platform and authenticate themselves with MetaMask, which makes it simpler to confirm their identity and ability to vote.

The block chain platform selected to implement the voting method is Ethereum. It enables decentralized apps (DApps) such as this voting system and facilitates the execution of smart contracts. The integrity of the voting process is improved by Ethereum's transparent and safe architecture, which guarantees that the votes and election data are unchangeable and publicly verifiable.

A. System Architecture

The block chain-based voting system's system design is based on a decentralized paradigm, with key components interacting with the Ethereum block chain to provide immutability, security, and transparency. The Front-End, Smart Contract, and Block chain layers are the three main layers of the architecture. Through a web interface made with HTML, CSS, and JavaScript, the Front-End communicates with users. Voters can view candidates and cast their ballots, and administrators can register both candidates and voters. The logic for managing voter registration, candidate registration, and vote casting is contained in the Solidity-written Smart Contract Layer (Fig. 1). By being implemented on the Ethereum block chain, which is the Block Chain Layer, these smart contracts guarantee that every transaction—including votes and registrations—is documented in an unchangeable and transparent ledger.

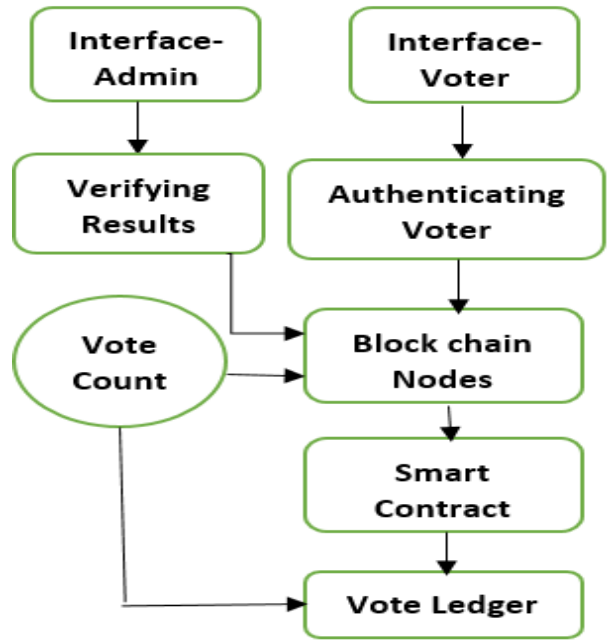


Fig. 1. System Architecture

The architecture incorporates MetaMask, a browser extension that functions as a digital wallet and enables safe communication between users and the Ethereum block chain. MetaMask offers authentication upon voter and administrator login, guaranteeing that only authenticated users are able to engage with the system. The Ethereum block chain is accessed by the front-end via Web3.js, which allows web pages to communicate with smart contracts (Fig. 2) and receive information like the current vote count or the list of candidates. The integrity along with transparency of the voting process are ensured by this architecture, which makes sure the voting system is secure and decentralized, preventing any central authority from changing the results or manipulating the data.

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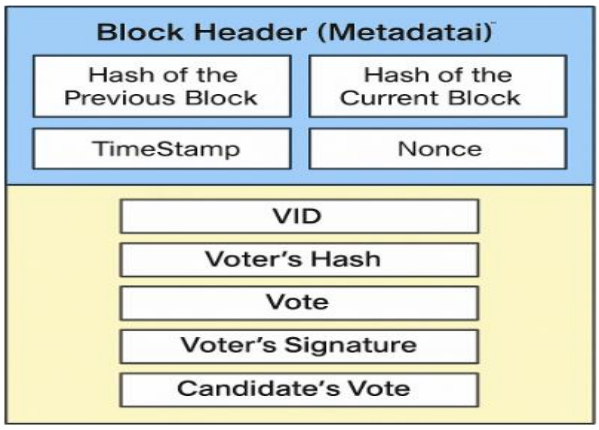


Fig. 2. Block Structure

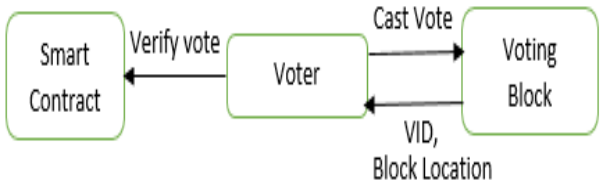


Fig. 3. Verification of Vote

B. *Operations of the Admin Dashboard*

Voter registration and candidate inclusion are handled by the admin dashboard. The addCandidate(name) function in the smart contract allows the admin to simply enter candidate data using the admin.html interface, which is then saved on the block chain. This guarantees the immutability and safe storage of candidate data (Fig. 3). By entering their wallet addresses in the admin.html interface, the administrator can also register voters. Each voter is granted authorization to vote by connecting to the block chain with the registerVoter(address) function. Only approved voters, as confirmed by their wallet addresses, will be able to cast ballots thanks to this measure.

C. *User Connection and Authentication*

Voters start by visiting the index.html homepage and logging into the decentralized application (DApp). To ensure that the voter's identity is confirmed, the system creates a secure connection between the DApp and MetaMask using the connect () function. Voters authenticate themselves by connecting their MetaMask wallet, and the wallet address is compared to the list of registered voters. Voting is only available to those whose addresses the administrator has registered. To assure that only the genuine eligible, registered voters can be indulged in the election process, this step is essential.

D. *Viewing Candidates and Casting Votes*

Voters can examine a list of all candidates on the voting.html portal after successfully authenticating. For the purpose of keeping it current and impenetrable, this list is dynamically extracted from the block chain. After choosing their favorite candidate, voters can cast their ballots. Voters choose a candidate and then confirm their selection in a straightforward and secure voting process. After confirmation, the vote is recorded on the block chain by calling the vote(candidateId) function. This guarantees that the vote is safely and permanently recorded, guarding against any kind of manipulation or meddling.

E. *Validation and Storage of Transactions*

The smart contract verifies the legitimacy of the vote and makes sure no voter has cast more than one ballot. Only legitimate votes are counted because the transaction is verified on the block chain. Decentralization and immutability are two of the block chain's intrinsic qualities that make it perfect for protecting vote data. After being verified, the vote is saved on the block chain, creating an unchangeable, transparent record of the election procedure. This technique ensures election fairness and transparency by preventing illegal alterations to the vote total.

F. *Compilation and Display of Results*

The results can be aggregated and shown instantly after the voting period is over. Vote data may be instantly retrieved thanks to the block chain, guaranteeing transparent and easily available results. The final results can be viewed by the administrator or authorized users via a specific result display interface or the admin dashboard. The final count is safe, precise, and unchangeable because every vote is recorded on the block chain. This guarantees that the election results accurately represent the outcome of the voting process and maintains its integrity.

V. PROPOSED METHODOLOGY

This project's block chain-based voting system effectively guarantees voting procedures' immutability, security, and transparency. Extensive testing was conducted to verify the system's functionality, and the usage of Ethereum smart contracts made it possible to cast votes and get results with ease. With MetaMask, voters could safely cast their ballots, and the administrator could quickly add candidates and register voters (Fig. 4). Votes could not be changed once they were cast since the block chain was decentralized, making it impossible to tamper with them.

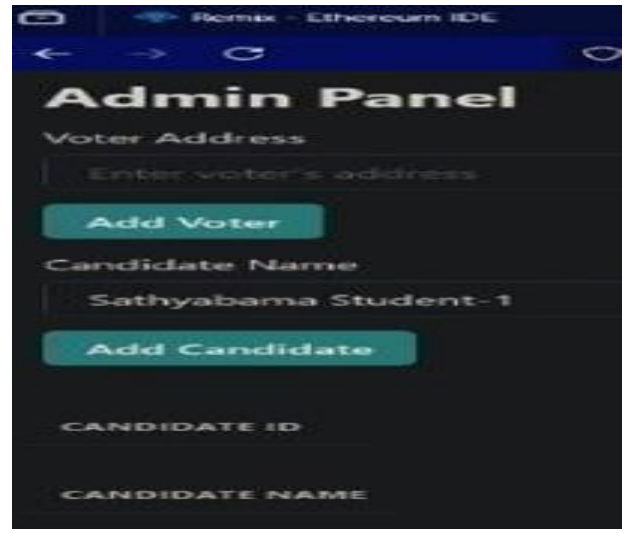


Fig. 4. Admin Panel

While the system performed OK while managing several transactions at once, larger elections would benefit from scalability enhancements which is shown in Fig. 5, where the

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registration is done using Metamask. By choosing Proof of Authority as the consensus technique, the system became more effective while preserving decentralization by striking a balance between speed and security.

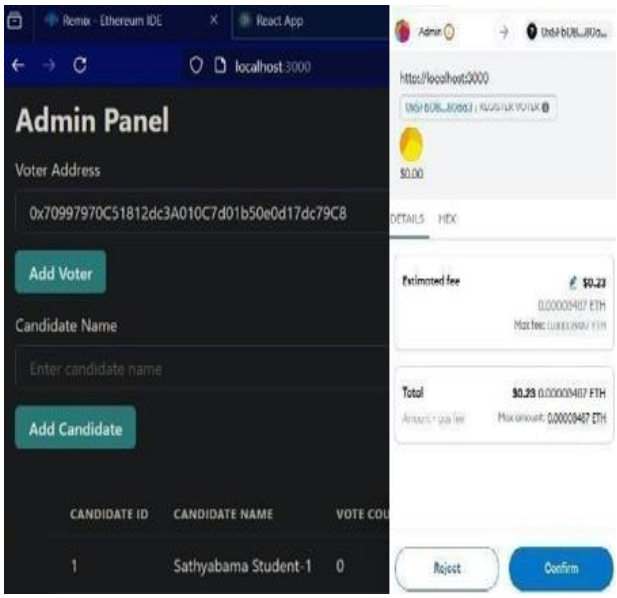


Fig. 5. Registering of Candidate Using Metamask

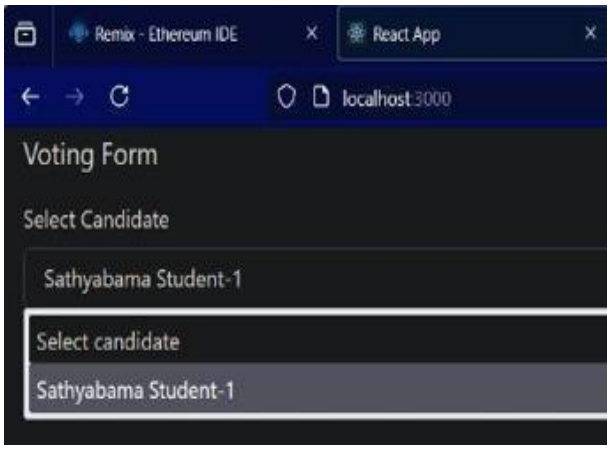


Fig. 6. Candidate Selection

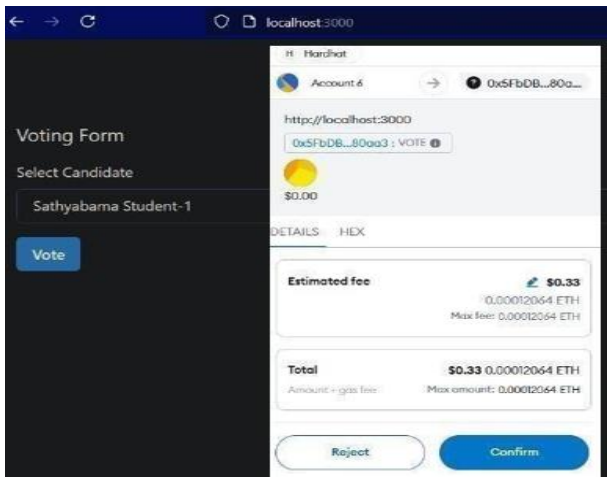


Fig. 7. Voting Done by Using Metamask Wallet

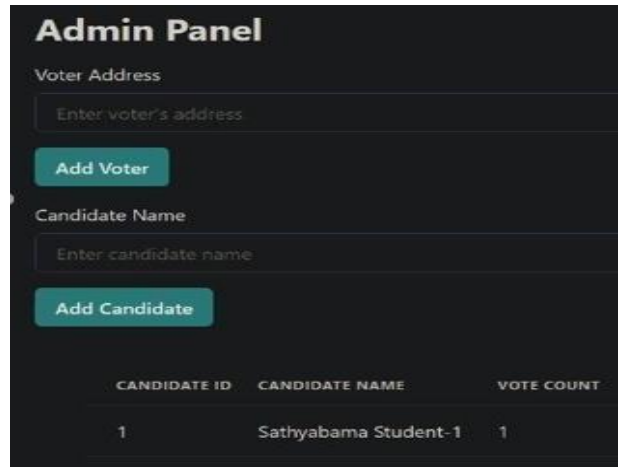


Fig. 8. Voting Result

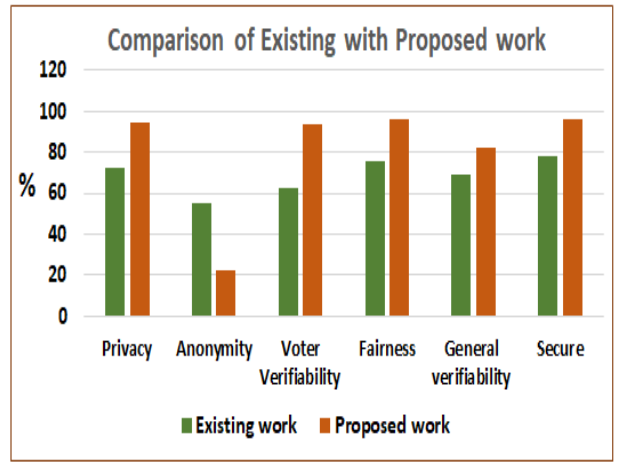


Fig. 9. Comparison of Existing with Proposed Work

Fig. 6 displays the Candidate Selection and Fig. 7 depicts the example of a Voting which is done by using Metamask Wallet. Fig. 8 shows the results of successful voting. Fig. 9 depicts the comparative study of Proposed System with the Existing System. The Proposed work excelled well by various factors when compared with the existing methods. Nevertheless, there were certain difficulties, especially with regard to the Ethereum network's transaction fees. Gas prices varied according on network congestion, which might have limited the system's viability for major elections. Although the UI was easy to use, non-technical consumers could benefit from more enhancements. Notwithstanding these difficulties, the system showed promise in digital voting, underscoring block chain's ability to solve problems like election fraud and vote tampering that are frequent in conventional voting systems.

VI. Conclusion

In conclusion, the block chain-based voting system created for this project offers an election-conducting technique that is safe, open, and impenetrable. The method makes sure that votes are recorded and counted honestly, free from the possibility of manipulation, by utilizing smart contracts and decentralized block chain technology. The project offers a creative answer for next elections and shows

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how block chain technology has the ability to completely transform voting methods. It might be a good choice for bigger elections with more scalability and transaction cost improvements.

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